



AI-Powered Collections Strategy



Leveraging Agentic AI for Scalable, Fair, and
Effective Debt Management at Geldium

Proposed Future-State Framework



System Workflow: Inputs & Decision Logic

1 Inputs

Proposed feed from the validated predictive model:
risk probability + Low / Moderate / High tier

Underlying signals: Missed_Payments,
Credit_Utilization, Credit_Score, Debt-to-Income Ratio

Designed to refresh as new payment and account
activity becomes available, subject to model validation.

Account context: tenure, employment status, prior
outreach / hardship history

2 Decision Logic

Combines the model's risk tier with business rules
(contact-frequency caps, active hardship-plan status)

Applies the optimized classification threshold selected
during model evaluation to convert predicted
probability into an action tier.

Moderate/High-risk and borderline cases are routed
for review rather than auto-resolved

Rules + model work together – not model output alone
– to pick the next step

System Workflow: Actions & Learning



Actions

Automated SMS / email payment reminders for Low–Moderate risk accounts

Recommend a short-term hardship or debt-management plan for High-risk accounts, subject to Collections-agent approval.

Escalation to a Collections agent for review and approval when required

Learning Loop

Tracks repayment rate, customer engagement, and cost of collections per action taken

Feeds observed outcomes back into future threshold and business-rule reviews, subject to validation, fairness checks, and human oversight.

Re-audited on a regular cadence for accuracy and fairness (see Guardrails)

Role of Agentic AI

Autonomous

Send automated payment reminders (SMS/email) to Low–Moderate risk accounts

Choose contact channel and timing based on past customer engagement

Adjust outreach cadence based on repayment response

Flag accounts that cross a risk threshold for the next workflow step

Human Oversight

Approve hardship or debt-management plan offers

Review High-risk and borderline / near-threshold cases

Handle customer disputes or appeals

Approve actions with regulatory impact (credit-limit or account-status changes)

Responsible AI Guardrails

- 1 Explainability** — Where applicable, automated risk flags can include SHAP-based reason codes (e.g., high utilization or repeated missed payments) so that model-driven actions can be explained in plain language.
- 2 Bias monitoring** — The Fairness Audit from the Model Plan re-runs on a regular cadence, comparing outcomes across Location, Employment_Status, Age, and Card_Type, with alerts on disparities.
- 3 Human-in-the-loop** — Hardship offers, credit-limit changes, and High-risk cases always route to a Collections agent for review – never fully automated.
- 4 Compliance & audit trail** — Every decision (inputs, model score, action taken) is logged for regulatory review, and customers can appeal or request human review of any outcome.

Expected Business Impact

Business KPIs

10–15% relative reduction in 30/60/90-day delinquency (proposed pilot target, subject to validation)

Lower cost-per-collection vs. manual case handling

Higher hardship-plan enrollment before accounts default

Faster time-to-outreach via real-time flagging vs. periodic manual review

Customer Outcomes

Earlier, supportive contact instead of late-stage collections pressure

Consistent, explainable treatment across customer segments

Clear channel to appeal or request a human review

Scalable support that reaches the full portfolio, not just manually reviewed accounts